

Assessment Test – Software Tester (QA)

1. Test Case Design

A. User Registration

TC ID	Test Scenario / Test Case	Steps	Expected Result	Type
REG-01	Register with valid details	Enter valid name, email and password, then submit	User account should be created successfully	Positive
REG-02	Register with all fields empty	Leave all fields blank and submit	Required field validation messages should be displayed	Negative
REG-03	Register with empty name	Leave name blank and enter other valid details	System should display name validation	Negative
REG-04	Register with invalid email	Enter an invalid email format	System should reject the email and show validation message	Negative
REG-05	Register with already registered email	Enter an email that already exists	System should prevent duplicate registration	Negative
REG-06	Register with invalid/weak password	Enter a password that does not meet password requirements	System should display appropriate password validation	Negative
REG-07	Register with minimum valid password	Enter password at the minimum allowed length	Registration should succeed if all other details are valid	Edge
REG-08	Register with very long input	Enter extremely long values in input fields	Application should safely reject or handle the input	Edge
REG-09	Enter special characters in input	Enter supported and unsupported special characters	Application should handle input safely and correctly	Edge
REG-10	Submit registration multiple times	Click Register repeatedly	Only one account should be created	Edge

B. User Login

TC ID	Test Scenario /	Steps	Expected	Type
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	Test Case		Result	
LOGIN-01	Login with valid credentials	Enter registered email and correct password	User should be logged in successfully	Positive
LOGIN-02	Login with incorrect password	Enter valid email and incorrect password	Login should fail with an appropriate error message	Negative
LOGIN-03	Login with unregistered email	Enter an email that is not registered	Login should fail	Negative
LOGIN-04	Login with empty fields	Leave email and password blank and submit	Required field validation should be displayed	Negative
LOGIN-05	Login with invalid email format	Enter an invalid email format	Email validation should be displayed	Negative
LOGIN-06	Verify password masking	Enter password in password field	Password should be hidden/masked	Positive
LOGIN-07	Login with incorrect case	Change the case of email/password where applicable	System should handle credentials according to defined rules	Edge
LOGIN-08	Multiple failed login attempts	Enter incorrect credentials repeatedly	System should handle repeated attempts securely	Edge
LOGIN-09	Logout and access task page	Login, logout and try to access task page	User should not access protected pages after logout	Positive
LOGIN-10	Session handling	Login and remain inactive for a period	Session should behave according to application's security requirements	Edge

2. Task CRUD Operations

A. Create Task

TC ID	Test Scenario / Test Case	Steps	Expected Result	Type
CREATE-01	Create task with valid details	Enter valid task information and click Create	Task should be created successfully	Positive
CREATE-02	Create task with required field empty	Leave required task information blank and submit	Validation message should be displayed	Negative

CREATE-03	Create task with minimum valid input	Enter minimum accepted task information	Task should be created successfully	Edge
CREATE-04	Create task with maximum allowed input	Enter data up to the allowed limit	Task should be created successfully	Edge
CREATE-05	Create task with excessive input	Enter data beyond the allowed limit	Application should reject or safely handle the input	Negative
CREATE-06	Create duplicate task	Create the same task more than once	Application should handle duplicate tasks according to requirements	Edge
CREATE-07	Cancel task creation	Enter task details and cancel the operation	Task should not be saved	Positive
CREATE-08	Verify task after creation	Create a task and refresh the task list	Created task should remain available	Positive

B. View Task List

TC ID	Test Scenario / Test Case	Steps	Expected Result	Type
VIEW-01	View existing tasks	Login and open task list	Existing tasks should be displayed	Positive
VIEW-02	View task list with no tasks	Login with a user who has no tasks	Appropriate empty-state message should be displayed	Edge
VIEW-03	Verify newly created task	Create a task and open the task list	Newly created task should appear	Positive
VIEW-04	Verify task information	Compare displayed task details with saved details	Correct task information should be displayed	Positive
VIEW-05	View large number of tasks	Add multiple tasks and open task list	Tasks should load correctly without data loss	Edge
VIEW-06	Refresh task list	Refresh the task-list page	Task data should remain consistent	Positive

C. Edit Task

TC ID	Test Scenario / Test Case	Steps	Expected Result	Type
EDIT-01	Edit an existing task	Select a task, modify its details and save	Task should be updated successfully	Positive
EDIT-02	Edit task with	Remove	Validation	Negative

	empty required field	required information and save	message should be displayed	
EDIT-03	Edit task with invalid input	Enter invalid task information and save	Invalid information should be rejected	Negative
EDIT-04	Edit with maximum valid input	Update task using maximum allowed input	Task should be updated successfully	Edge
EDIT-05	Cancel edit	Modify task and cancel the operation	Original task information should remain unchanged	Positive
EDIT-06	Verify edit after refresh	Edit a task and refresh the page	Updated information should remain available	Positive
EDIT-07	Attempt unauthorized edit	Attempt to edit another user's task	System should prevent unauthorized modification	Negative

D. Delete Task

TC ID	Test Scenario / Test Case	Steps	Expected Result	Type
DELETE-01	Delete existing task	Select a task and click Delete	Task should be deleted successfully	Positive
DELETE-02	Cancel deletion	Click Delete and cancel the confirmation	Task should remain unchanged	Positive
DELETE-03	Confirm deletion	Click Delete and confirm	Task should be removed from the task list	Positive
DELETE-04	Verify deletion after refresh	Delete a task and refresh the page	Deleted task should not reappear	Positive
DELETE-05	Delete already deleted task	Attempt to delete a task that no longer exists	Application should handle the request gracefully	Edge
DELETE-06	Attempt unauthorized deletion	Attempt to delete another user's task	System should prevent unauthorized deletion	Negative

3. Input Validation & Error Handling

TC ID	Test Scenario / Test Case	Steps	Expected Result	Type
VAL-01	Required field validation	Submit a form with required fields empty	Clear validation messages should be displayed	Negative

VAL-02	Invalid email validation	Enter an incorrectly formatted email	Invalid email should be rejected	Negative
VAL-03	Minimum input validation	Enter minimum allowed values	Valid minimum values should be accepted	Edge
VAL-04	Maximum input validation	Enter maximum allowed values	Valid maximum values should be accepted	Edge
VAL-05	Excessively long input	Enter very long text	Application should safely reject or limit the input	Edge
VAL-06	Special character handling	Enter special characters in fields	Application should handle input safely	Edge
VAL-07	Unexpected/unsafe input	Enter unexpected or potentially unsafe input	Input should be handled safely without application failure	Negative
VAL-08	Database/server error	Perform an operation while backend/database is unavailable	User should receive a meaningful error instead of an application crash	Negative
VAL-09	Network interruption during save	Interrupt connection while saving a task	Application should handle failure without incorrect/duplicate data	Edge
VAL-10	Multiple submit clicks	Quickly click Create/Submit multiple times	Duplicate records should not be created	Edge

4. Potential Bugs / Risk Areas

#	Potential Bug / Risk Area	Severity	Reason / Impact
1	Incorrect credentials may be accepted during login	Critical	Could allow unauthorized users to access an account and its task data
2	A user may be able to view another user's tasks	Critical	Creates a serious privacy and authorization issue
3	A user may be able to edit or delete another user's task	Critical	Unauthorized modification or deletion could cause data loss
4	Duplicate user accounts may be created using the same email	Major	Can create inconsistent user records and login/account-management problems

5	Deleted tasks may reappear after refreshing the page	Major	Indicates deletion may not be correctly persisted in the database
6	Edited task information may not be saved correctly	Major	Users could lose important changes and continue seeing outdated information
7	Multiple clicks on Create may create duplicate tasks	Major	Can create duplicate database records and inaccurate task lists
8	Application may crash when unexpected or invalid input is submitted	Major	Poor error handling can make the application unstable or unusable
9	Database/server errors may expose technical information	Major	Could reveal sensitive system or database information and create a security risk
10	Task list may display incorrect or outdated data	Minor	Users may make decisions based on incorrect task information

5. Testing Approach

Testing objective: Verify that the Task Management Application works correctly, handles invalid inputs safely, protects user data, and maintains consistency between the application and database.

Priority 1 – Authentication & Authorization

- Registration with valid and invalid information
- Login with valid and invalid credentials
- Logout and session handling
- Unauthorized access to another user's tasks

Priority 2 – Task CRUD

- Create task
- View task list
- Edit task
- Delete task
- Verify changes are correctly stored in the database

Priority 3 – Validation & Error Handling

- Required fields
- Invalid formats
- Minimum and maximum values
- Unexpected input

- Server/database errors
- Network failures

Priority 4 – Data Integrity

- Created tasks remain after refresh
- Edited tasks retain their changes
- Deleted tasks do not reappear
- Duplicate records are not created

6. Conclusion

The main objective of testing would be to ensure that all required features work correctly under normal conditions and that the application handles invalid, unexpected, and boundary inputs safely. Special attention should be given to authentication, authorization, database persistence, CRUD operations, validation, and error handling, as failures in these areas can have significant impact on users and application data.